

DIAGRAM 1: USE CASE ARCHITECTURAL DIAGRAM

UML 2.5 SPECIFICATION

VOTEPULSE E-VOTING SYSTEM -- ARCHITECTURAL SPECIFICATION

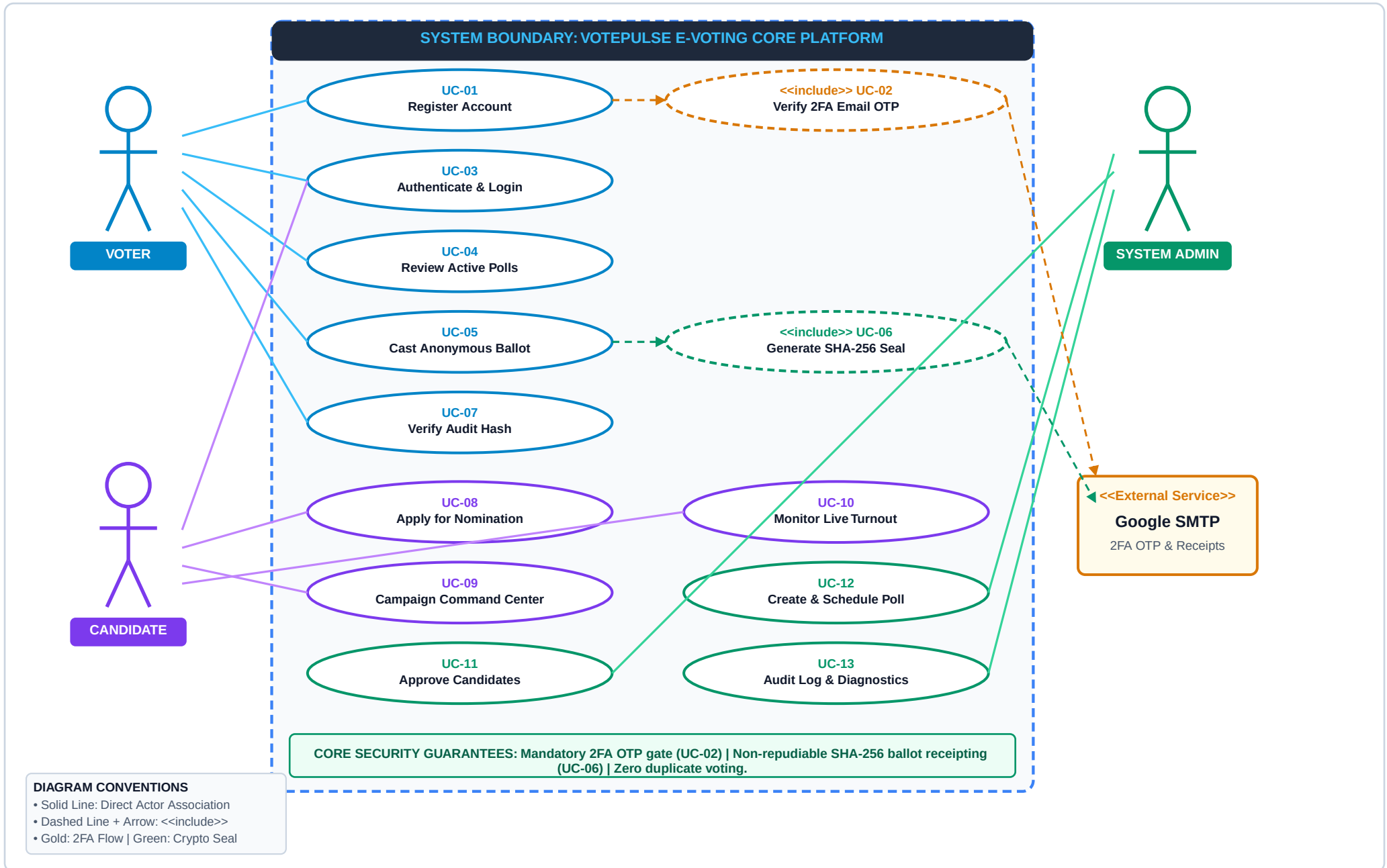


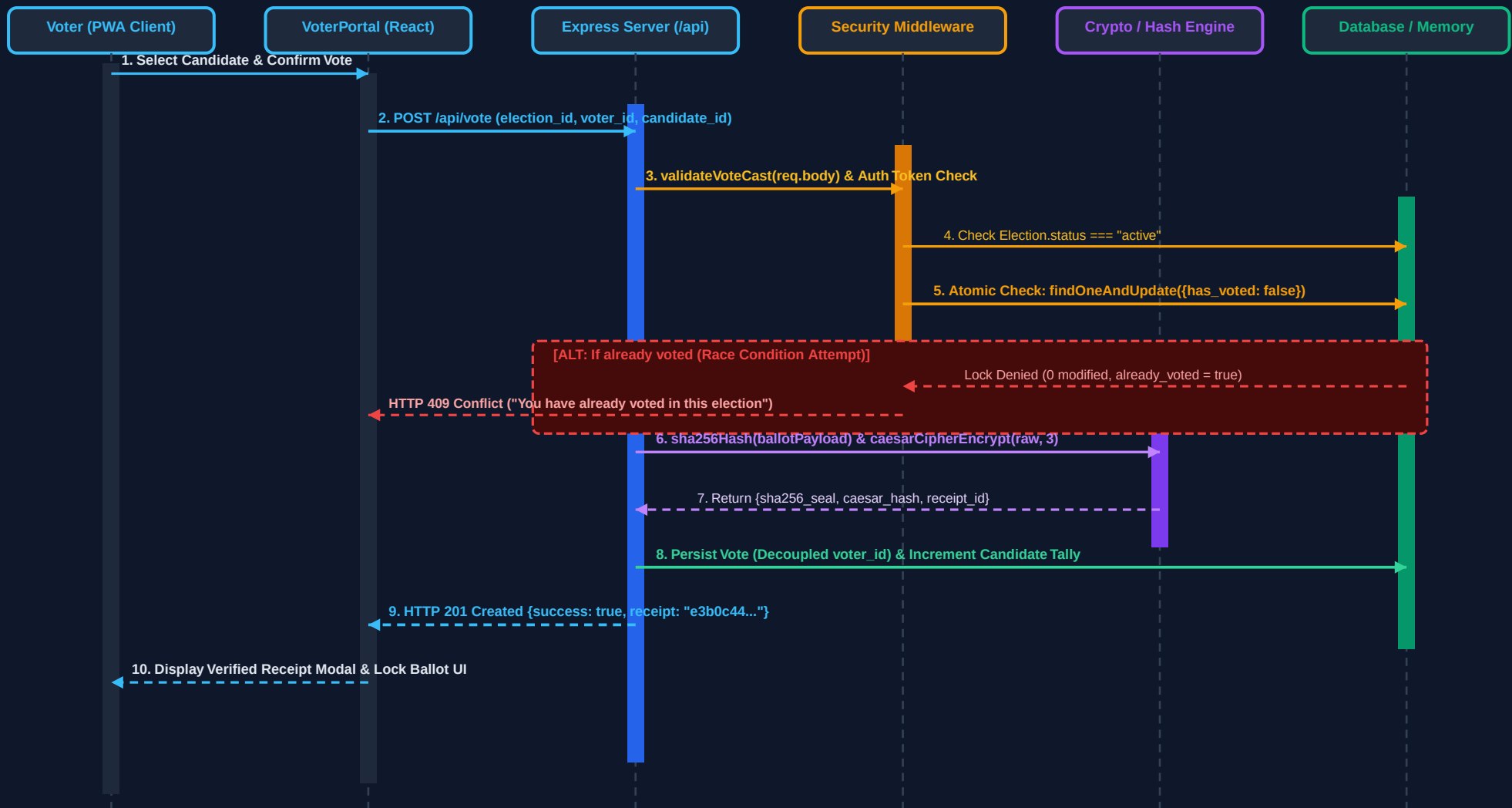
DIAGRAM 2: SEQUENCE DIAGRAM: BALLOT CASTING & SEALING

DYNAMIC INTERACTION MODEL

VOTEPULSE E-VOTING SYSTEM -- ARCHITECTURAL SPECIFICATION

VOTEPULSE: END-TO-END SECURE BALLOT CASTING SEQUENCE DIAGRAM

CONCURRENCY & CRYPTOGRAPHIC SEAL FLOW

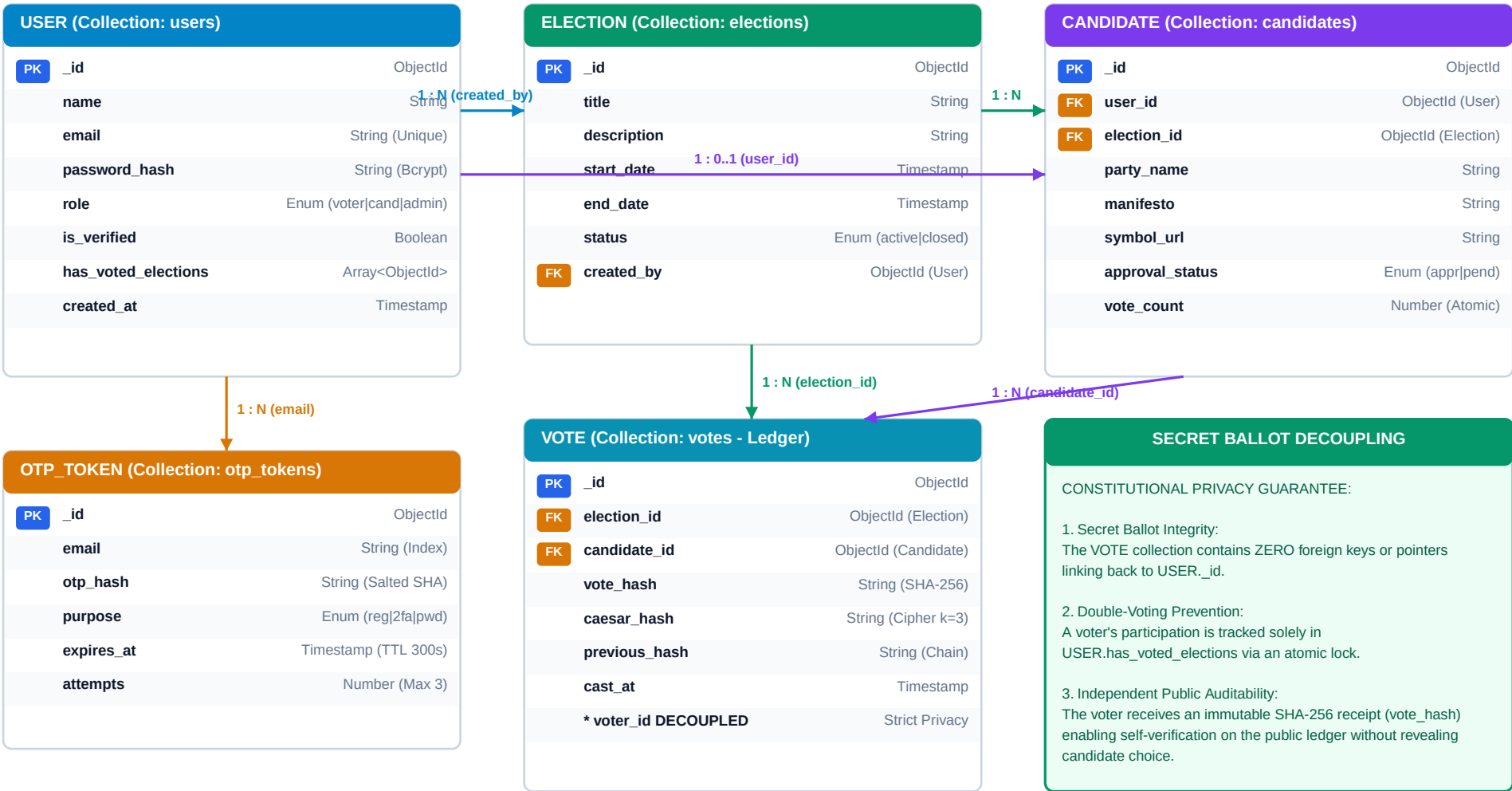


ZERO-TOLERANCE RACE DEFENSE: Native MongoDB atomic locks reject concurrent double-vote attempts in < 2ms without table locking.

DIAGRAM 3: ENTITY-RELATIONSHIP (ER) SCHEMA ARCHITECTURE

VOTEPULSE E-VOTING SYSTEM -- ARCHITECTURAL SPECIFICATION

DATA MODELING & PERSISTENCE

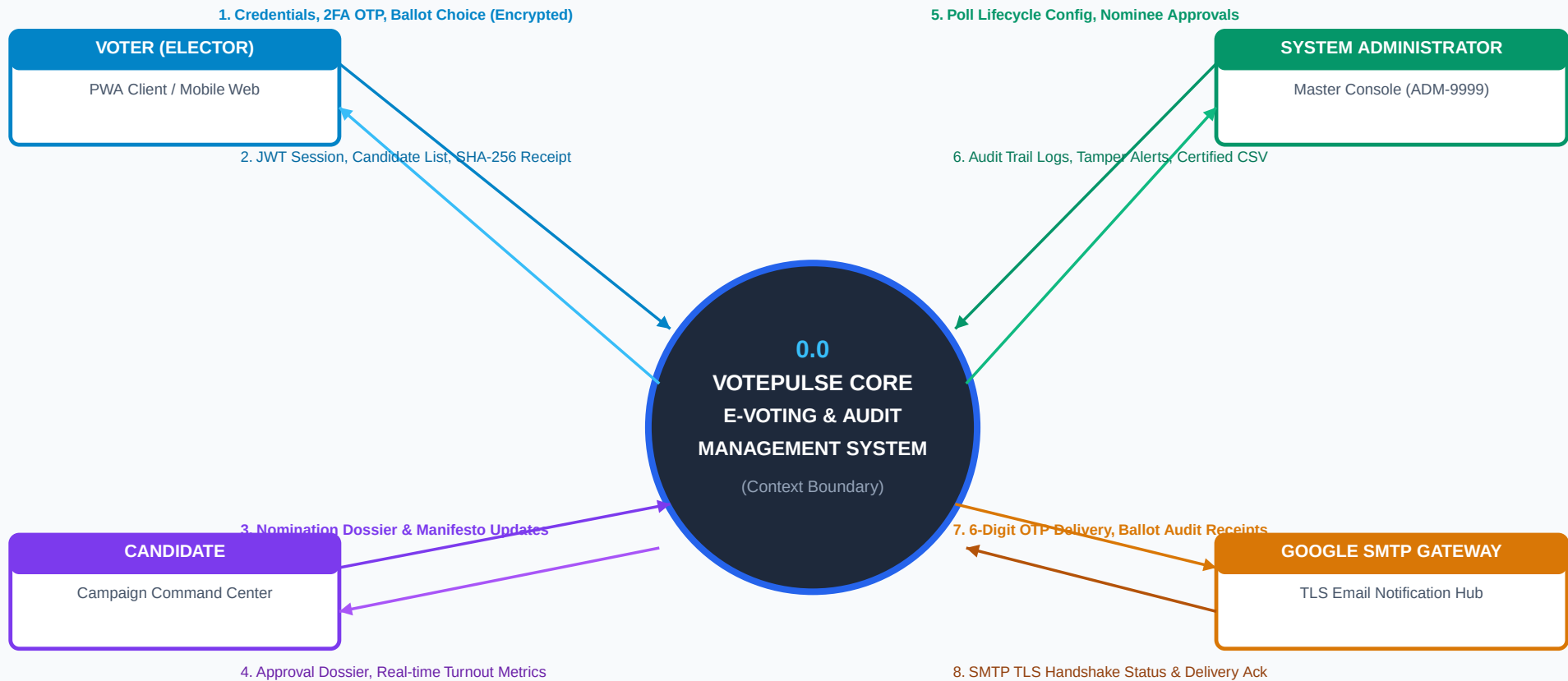


PRIMARY KEYS: [PK] | FOREIGN KEYS: [FK] | TTL AUTO-EXPIRY: 300 SECONDS | CRYPTOGRAPHIC BALLOT ANONYMIZATION

DIAGRAM 4: DATA FLOW DIAGRAM (DFD) LEVEL 0: CONTEXT MODEL

CONTEXT-LEVEL DATA FLOW

VOTEPULSE E-VOTING SYSTEM -- ARCHITECTURAL SPECIFICATION

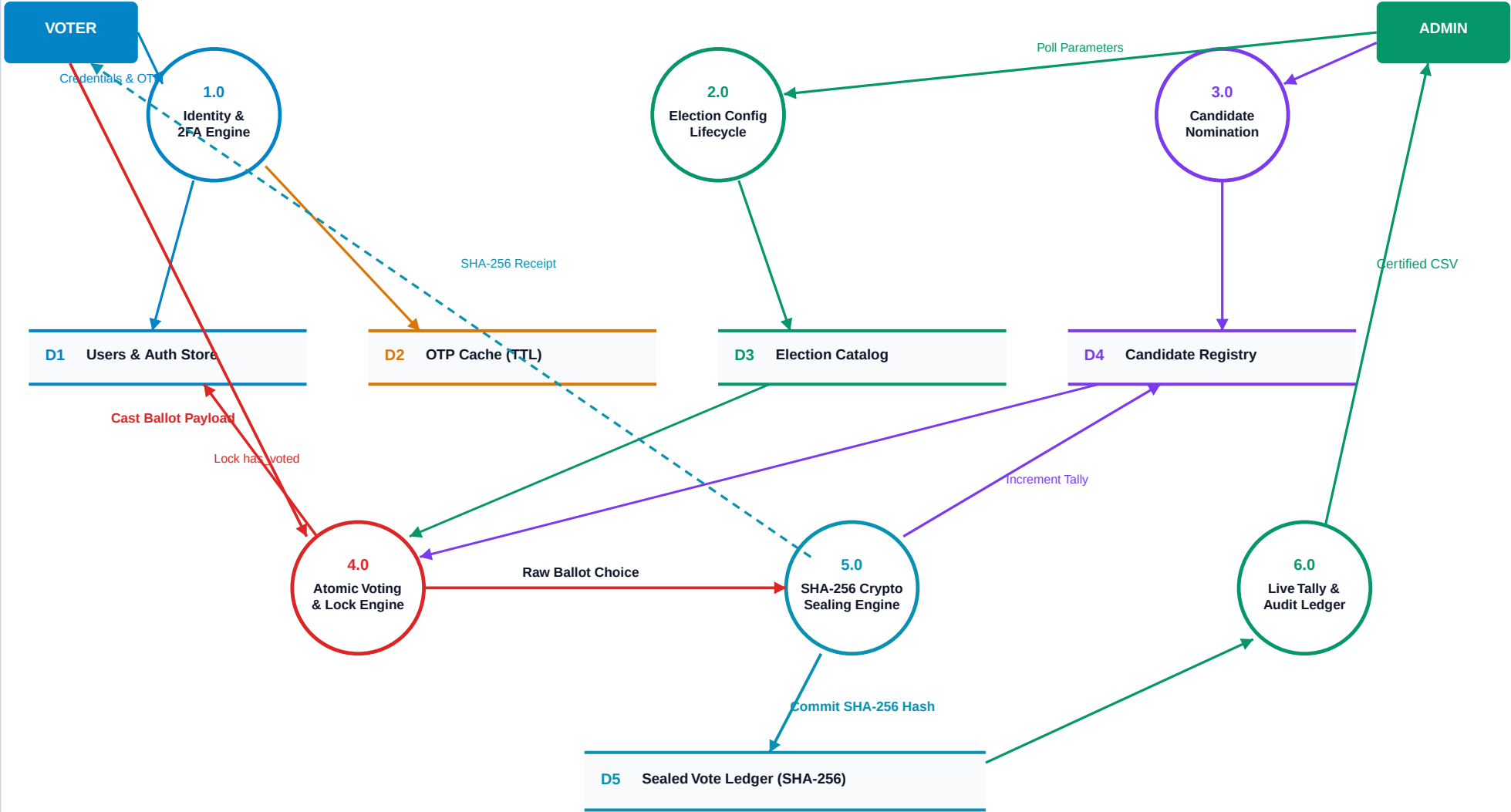


DFD LEVEL 0 ARCHITECTURE: Represents the black-box contextual model. All perimeter transmissions are encrypted via HTTPS / TLS 1.3.

DIAGRAM 5: DATA FLOW DIAGRAM (DFD) LEVEL 1: PROCESS DECOMPOSITION

DETAILED DATA FLOW MODEL

VOTEPULSE E-VOTING SYSTEM -- ARCHITECTURAL SPECIFICATION



DFD LEVEL 1 SPECIFICATION: Decoupled process decomposition ensures zero ballot trace to D1 and zero race conditions on D5.

